A

PROJECT REPORT

ON

"SMART GRID"

In Partial Fulfilment of the Requirement for the Award of

BACHELOR'S DEGREE IN
INSTRUMENTATION & CONTROL ENGINEERING

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OF
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ABSTRACT

Smart Grid is a large complex network connecting the sources of energy to the loads of the utilities in order to achieve the efficient use of energy with low energy costs, thus, to achieve this, the grid communicates with the sources and the utilities and reacts by making changes in the grid. The changes in the grid involves whether to use the conventional sources or non-conventional sources and synchronization of non-conventional sources with the conventional sources during the peak hours where energy requirement is high, efficiency of the conventional sources is low and energy costs are high. The changes to be made in the grid are configured by the intelligent control algorithm in the PLC in coordination with cloud analytics.

The cloud-analytics involves the prediction of the energy demands of the utilities, the peak hours and energy generation by the various sources and synchronizing this data with intelligent control algorithm and making changes in the grid accordingly. The smart grid also involves the fault detection systems in the generation, utilities and transmission and reacting to it smartly so the flow of energy is not interrupted in the grid in case a fault is detected. It also involves automatic billing system for the utilities on account of energy consumption. Also connection for utilities can be made off and on remotely from control room. Thus, the smart grid is a complex system which involves intelligent control and management of the energy flow in the grid, data analytics to assist the control algorithm, fault detection and automatic billing for the utilities.

Chapter 1

INTRODUCTION

Energy demand is increasing day by day because of increasing population. It has become necessity of the time to provide energy efficiently. This is only possible by introducing 'Real time monitoring and automated control' in existing power grid. This is popularly known as 'Smart Grid'.

'Smart Grid' is an electricity supply network that uses Internet of Things(IoT) to detect and react to local changes in the utility. It also includes automated fault detection and correction in the grid, automated billing of the utility and efficient utilization of conventional and non-conventional power plants.

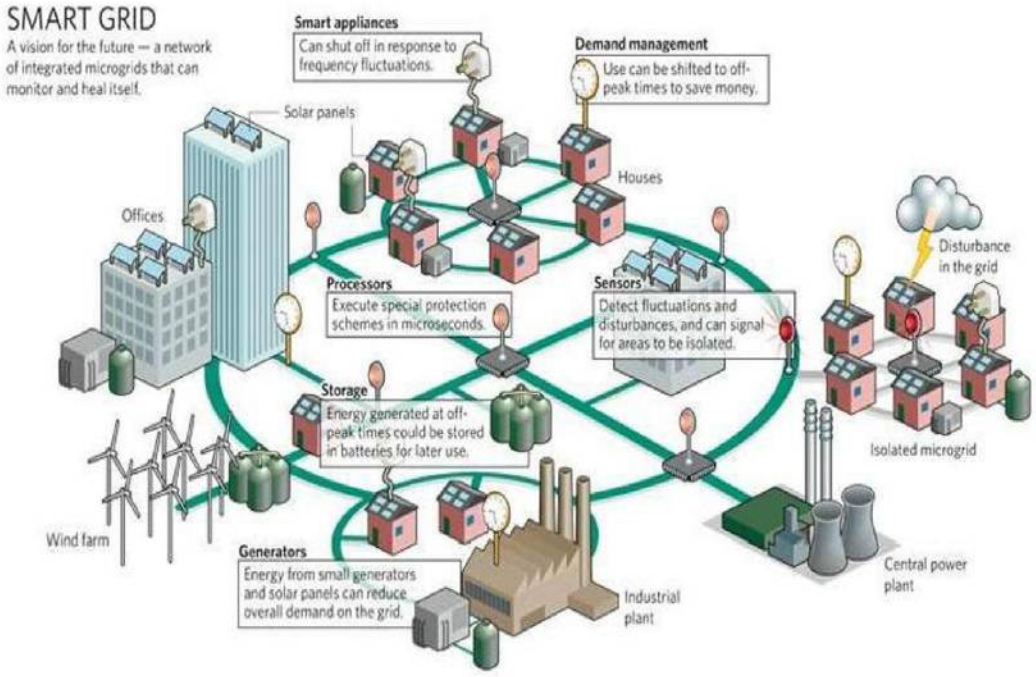


Figure 1.1: **Smart Grid Concept**

2.1 Traditional Grid

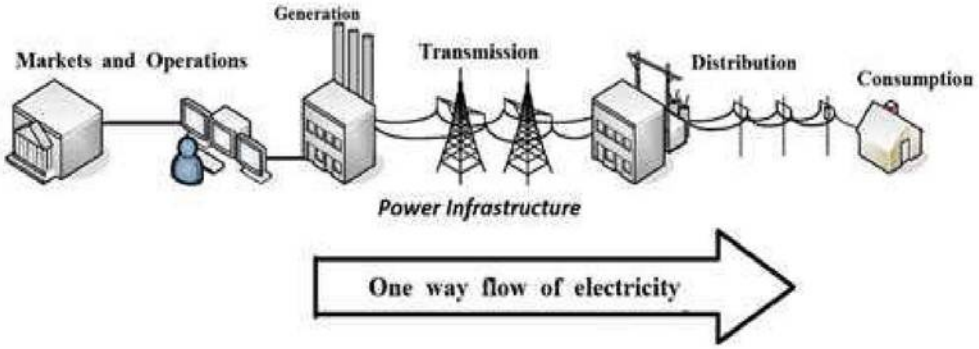


Figure 2.1: **Traditional Grid**

•The existing utility grid is a centralized system where power flows in one direction, from generation resources through the transmission-distribution system to the customer.

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•Centralized generation of electricity. Generation may or may not be located in the same geographic area as the load being served, which can often require transmission from distant locations.

•Existing utility grids may or may not include Supervisory Control and Data Acquisition (SCADA), sensors, computing, and communications to monitor grid performance.

•Utility systems may depend instead on separate reporting systems, periodic studies, and standalone outage management applications.

•Information to the customer is generally limited to a periodic bill for services consumed in a prior time period or billing cycle.

2.2 Smart Grid

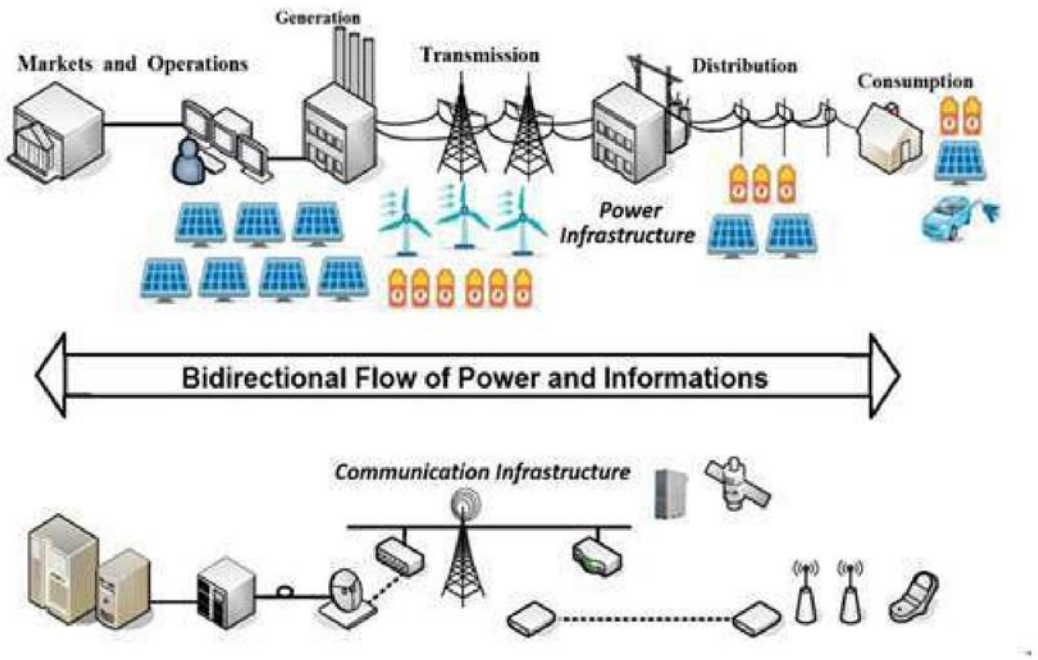


Figure 2.2: **Smart Grid**

•The smart grid is a decentralized system where power flows in both direction, from generation resources through the transmission-distribution system to the customer and vice-versa.

•Several generation options throughout the grid allows power sources to be located closer to their point of use, reducing investment in transmission and distribution, and in many cases reducing energy losses.

•Small generating resources are implemented that diversifies supply, reduces risks of major outages, and improves overall reliability.

•Sensors, remote monitoring, automated switches, reclosers, upgraded capacitor banks, and other equipment are integrated into the grid to provide end-to-end monitoring and control of the transmission and distribution network. Sensors provide the information for better understanding of grid operation, while control devices provide options for better management of system operation.

•Smart appliances with embedded price, event-sensing and energy management capability are implemented.

Chapter 3

Features of Smart Grid

The smart grid represents the full suite of current and proposed responses to the challenges of electricity supply. Because of the diverse range of factors there are numerous competing taxonomies and no agreement on a universal definition. Nevertheless, one possible categorization is given here.

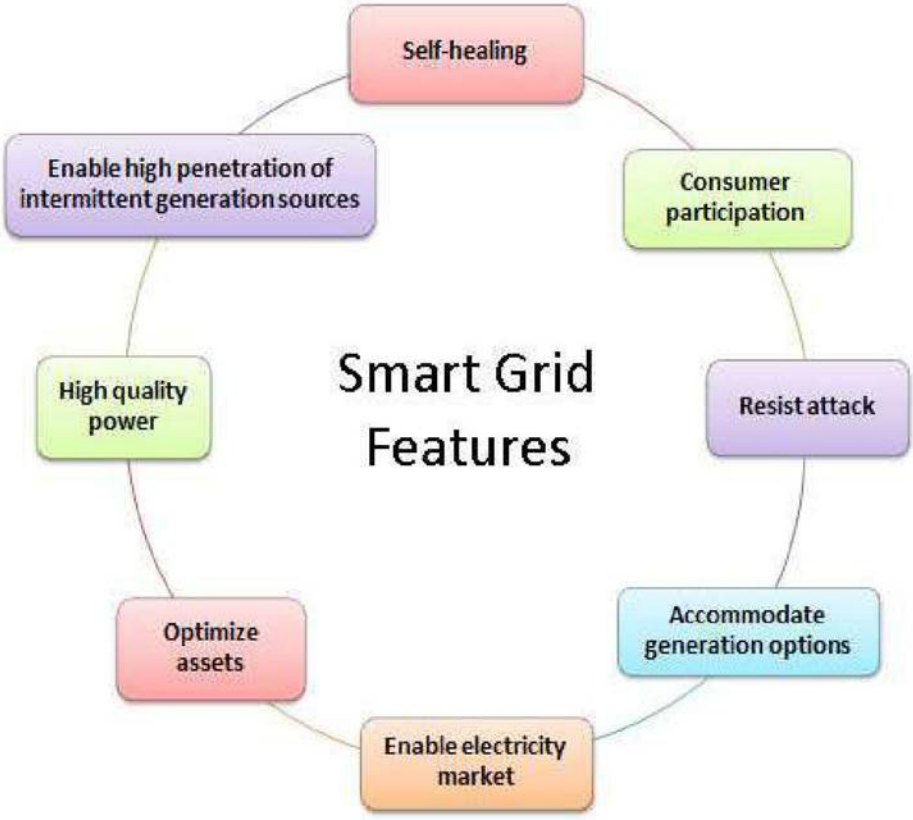


Figure 3.1: **Features of Smart Grid**

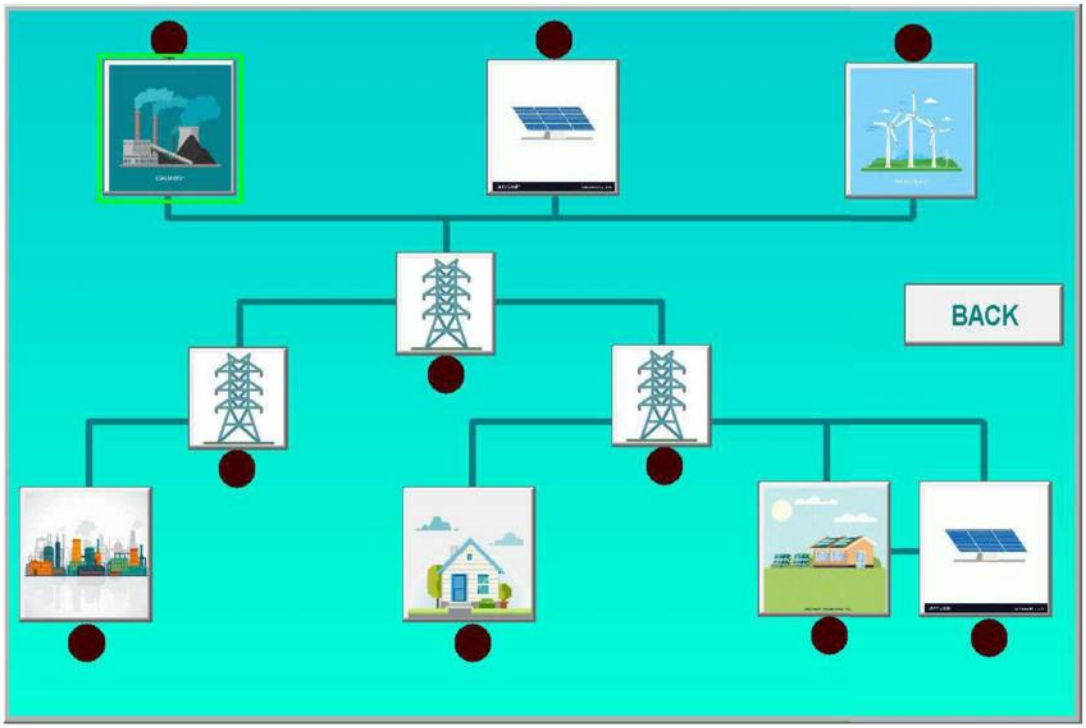


Figure 5.12: **SCADA Mimic Screen**

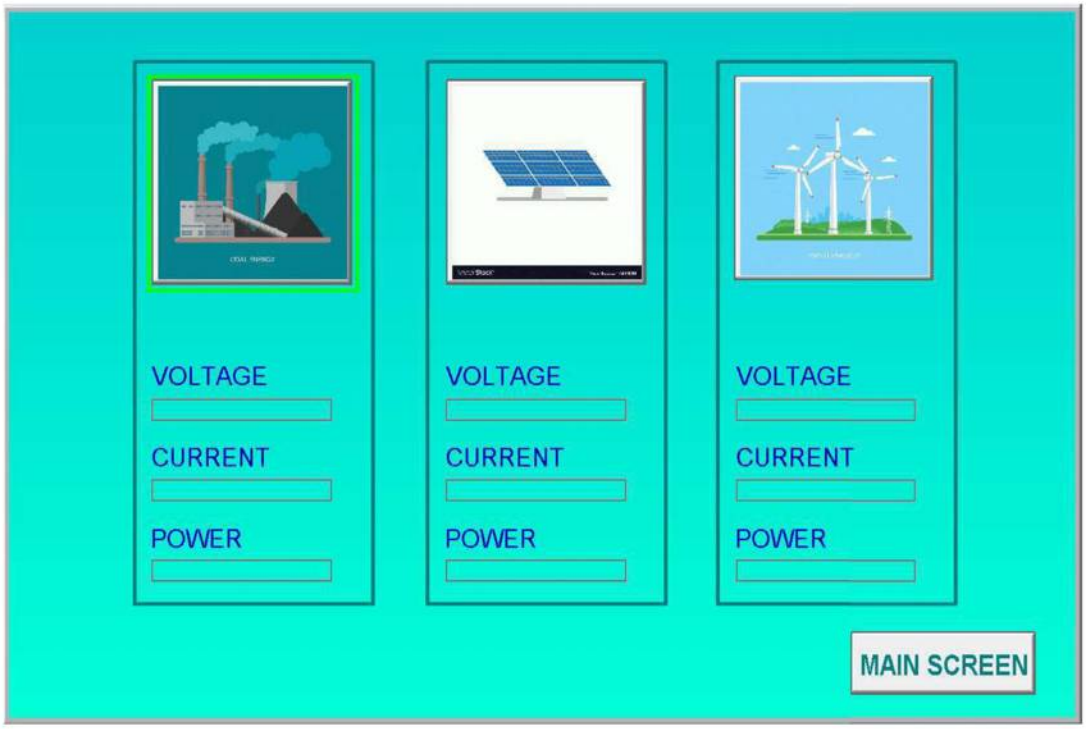


Figure 5.13: **SCADA Sources Screen**

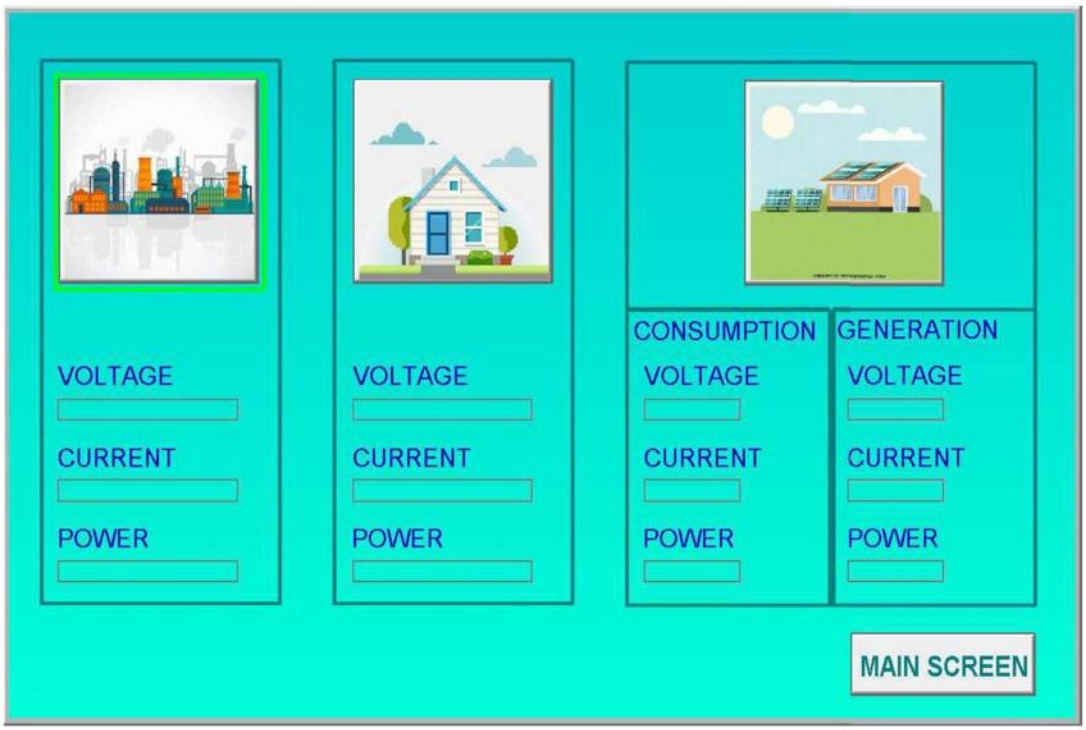


Figure 5.14: **SCADA Loads Screen**

During the condition when entire load on grid is lower, non contributing power stations can be disconnected from that grid which makes them available source for some other or parallel grid. This will efficiently use the power plant energy and save the cost of unused energy generation.

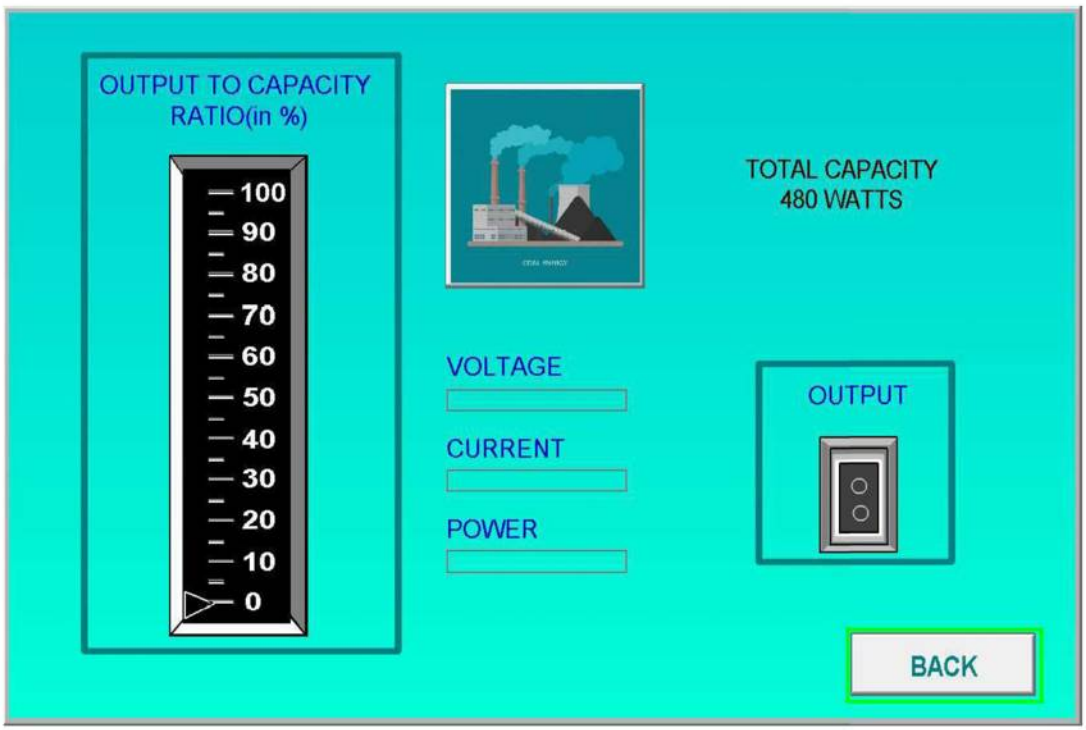


Figure 5.15: **SCADA Main all parameters monitoring Screen**

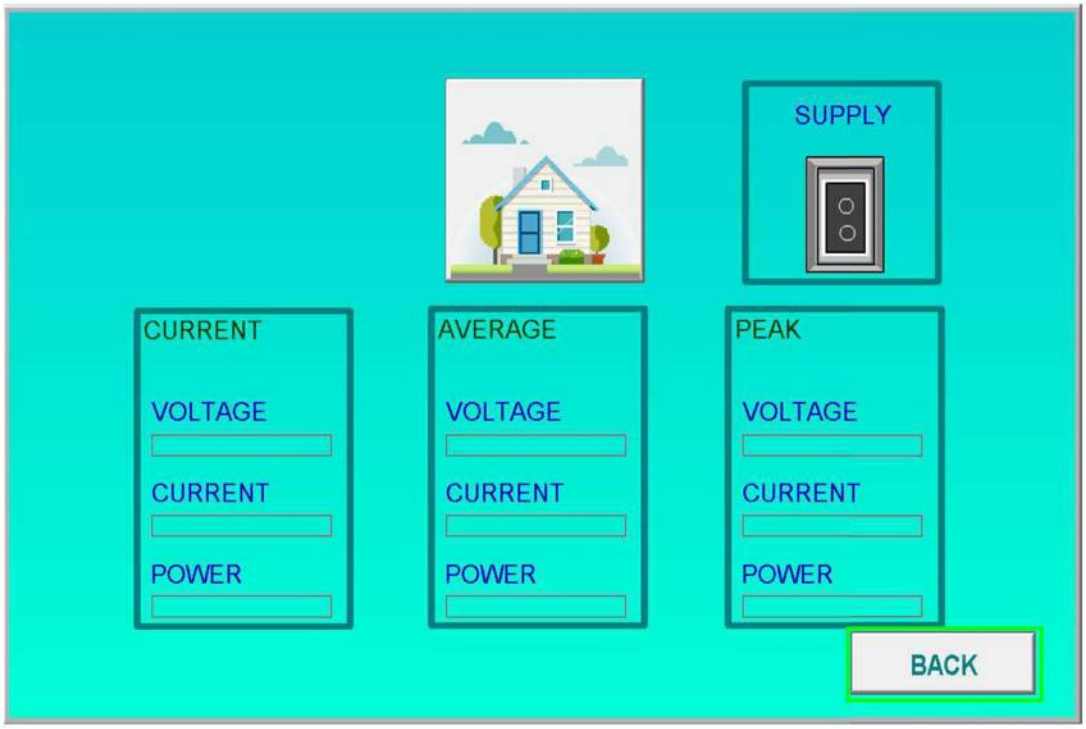
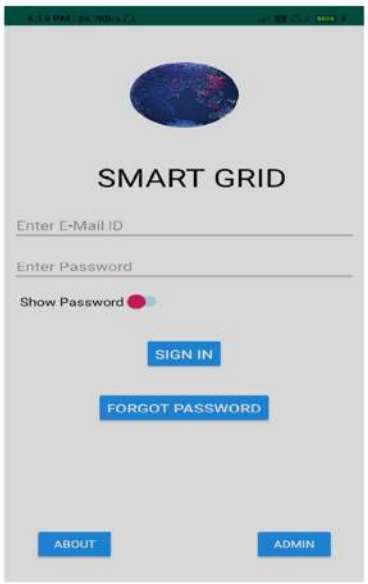


Figure 5.16: **SCADA Home all parameters monitoring Screen**



(a) **Login Screen**



(b) **Monitoring Screen**

Figure 5.18: **Application for Consumer**



(a) **Consumer Selection**



(b) **Monitoring Screen**

Figure 5.19: **Application for Authorized Personnel**

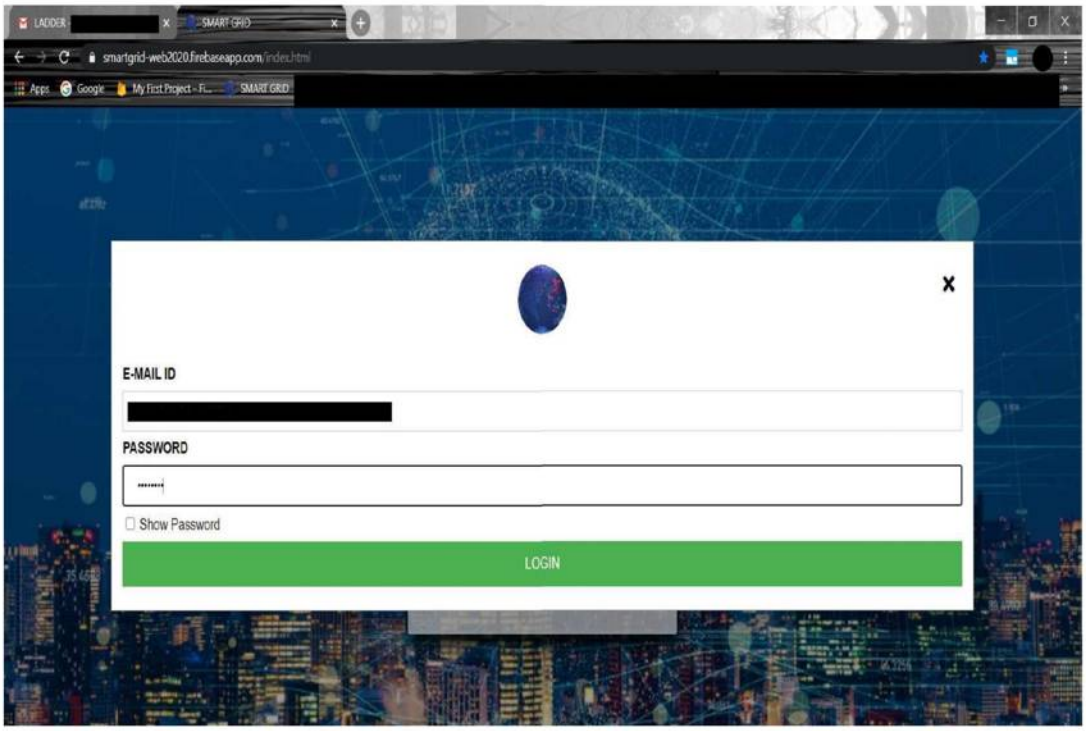


Figure 5.21: **Website Login Page**

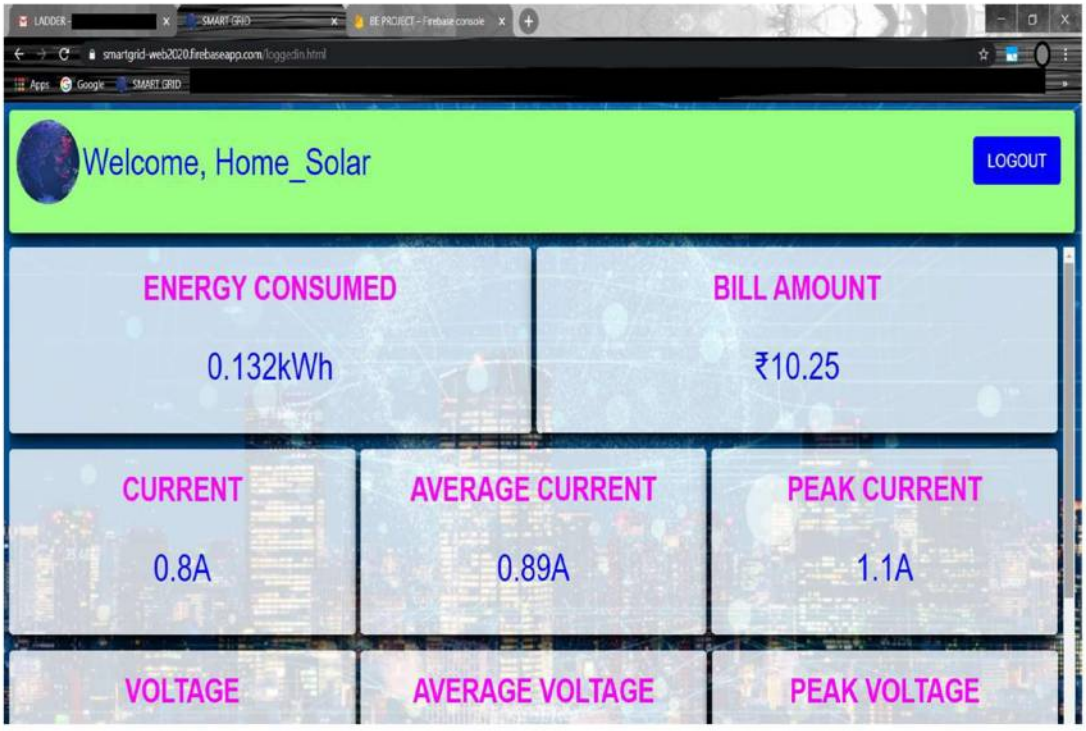


Figure 5.22: **Website Monitoring Page**

The Figure[5.22] shows the logged in window of customer with integrated solar panels.

6.1.2 Source Control Unit

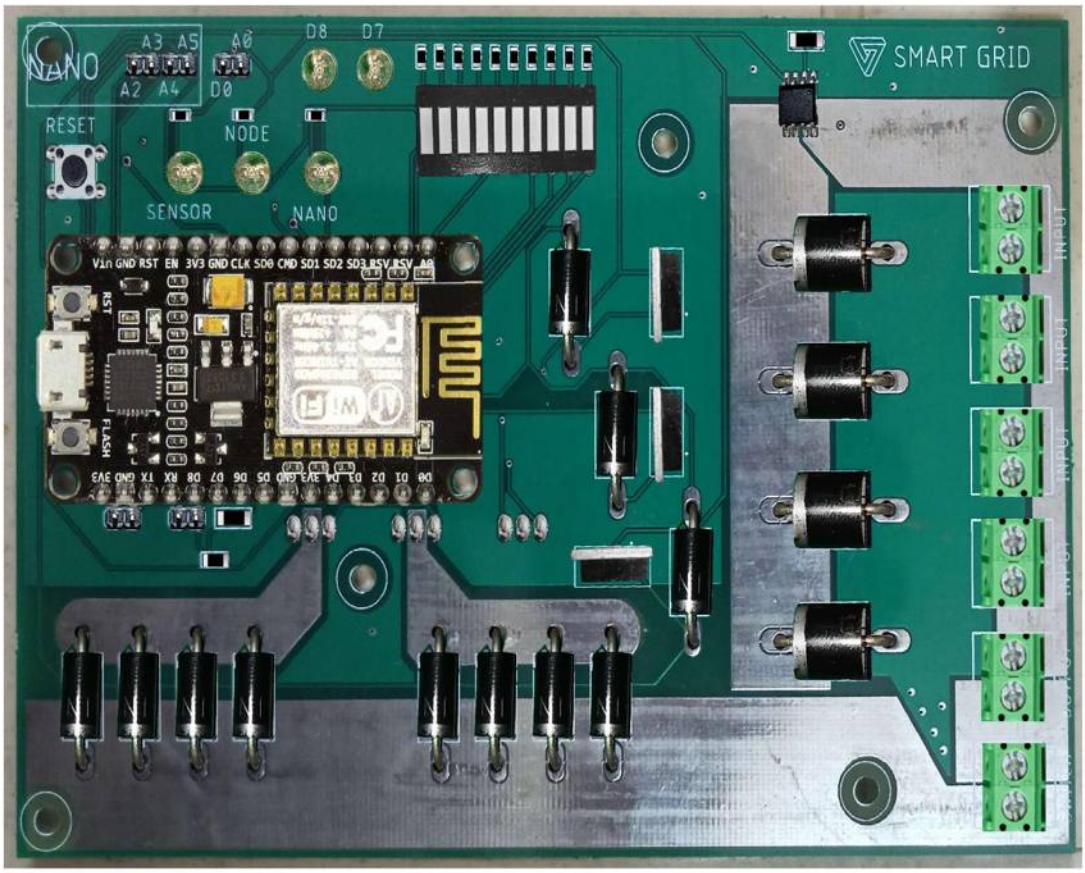


Figure 6.4: **Source Control Unit (Top Side)**

Based on design block diagram in Figure[6.1], PCB(Printed Circuit Board) was designed to arrange all the components on a single standalone board as in Figure[6.4]. As on this circuit board, the screw terminals at right are used to connect the power input from the SMPS mentioned earlier. Also the output power lines are taken through these screw terminals. The input and output power of this board is monitored and the nodeMcu in the circuit process the data from voltage and current sensor ACS712 and then transmits it wirelessly to the google firebase.

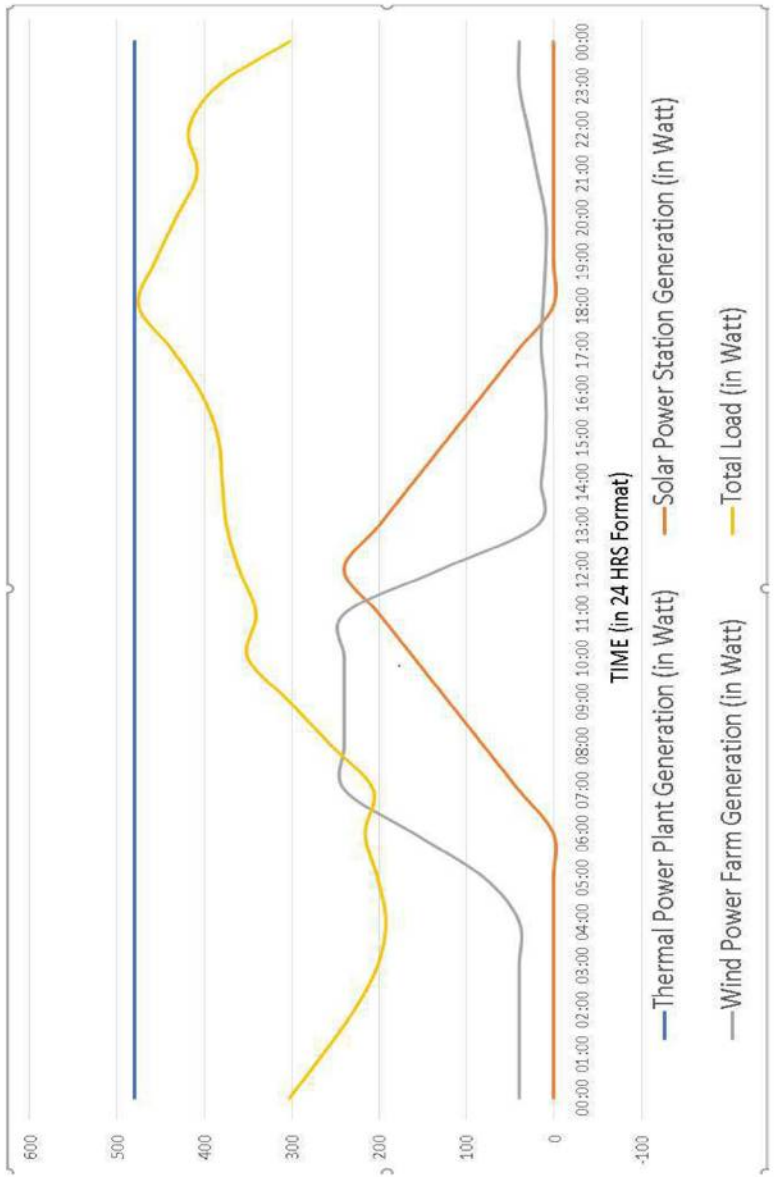


Figure 9.1: **Result 1**

Chapter 11

Conclusion & Future Scope

11.1 Conclusion

There is no doubt that the future belongs to the Smart Grid, and that power generation will change significantly by the time it becomes a reality.
Large power plants will continue to ensure the basic supply, but there will also be renewable energy sources, causing fluctuations in the grid. In the not too distant future, flexible intermediate storage of temporary excess power in the grid will be possible using electric vehicles and stationary storage units.
Sensors and smart meters will switch these units on or off, ensuring efficient load management.

11.2 Future Scope

The current mode of transmission and distribution of electricity has proven to be unreliable and inefficient. This is because the grid technology currently in use has changed very little since it was developed. Researchers are now experimenting with smart grid technologies to overcome the shortcomings of the traditional grid. A smart grid can help reduce greenhouse gas emissions by up to 211 million metric tons and is much more reliable than a traditional grid.